Appendix B

Answers to Selected Problems

Chapter 2

Problem 2.9.1: Specific weight=52.88 lb/ft^3. Specific gravity=0.8474.
Problem 2.9.2: Specific gravity at 85°F=0.845.
Problem 2.9.3: Specific gravity=0.7428.
Problem 2.9.7: Blended viscosity=145 SSU.
Problem 2.9.8:75% of liquid A and 25% of liquid B.

Chapter 3

Problem 3.15.1: Average velocity=8.03 ft/s. Reynolds number=203,038.
Problem 3.15.2: Darcy friction factor f=0.0164. Modified Colebrook-White friction factor=0.0167. Pressure drop=101.66 psi.
Problem 3.15.4: Total pressure=36.72 MPa. Four pump stations.

Chapter 4

Problem 4.5.2: MAOP=1380 psi. Hydrotest range=1725 to 1820 psi.
Problem 4.5.4: Volume per mile=926.19 bbl.

Chapter 5

Problem 5.13.1: (a) 914 psi. (b) BHP=2201; motor HP=2293 HP. (c) 2500 HP. (d) 6510 bbl/hr.
Problem 5.13.2:640 mm diameter, 10 mm wall thickness.
Problem 5.13.3:8177 bbl/hr gasoline flow rate, gasoline head=4277 ft; diesel head=3724 ft.

Chapter 6

Problem 6.7.1: (a) 0.375 in. wall thickness, (b) Two pump stations, (c) Two additional pump stations, (d) 6888 HP and 18,530 HP at 75% pump efficiency.

Chapter 7

Problem 7.14.1: (a) Trim impeller in pump A or pump B. (b) 9.2 in. impeller for pump A. (c) 3282 RPM.
Problem 7.14.2:

Q(gal/min):	0	500	1000	1500	2000
P(psi):	130	208	419	758	1224

Chapter 8

Problem 8.5.1: (a) 122 psi. (b) $111,542 based on 350 days operation/yr.
Problem 8.5.2: Speed of VSD pump=3126 RPM.

Chapter 9

Problem 9.4.1: Minimum flow rate=1050 bbl/hr; HP=298.
Problem 9.4.2: Minimum flow rate=1500 bbl/hr; HP=895.

Chapter 10

Problem 10.9.1: (a) R=212,750. (b) 3.84 in. throat diameter. (c) R=221,288.
Problem 10.9.3: Flow rate=959 gal/min.

Chapter 11

Problem 11.6.1: Potential surge=903 ft. Wave speed=3913 ft/s.
Problem 11.6.3:

Thin wall:	3967 ft/s	3926 ft/s	3868 ft/s
Thick wall:	3942 ft/s	3903 ft/s	3847 ft/s